Your submitted solution is only **valid if it contains the following paragraphs** at the top of the program. Please ***fill*** in ***your name*** and ***Neptun code***.

```
/*<NAME>, <NEPTUN>
This solution was prepared and submitted by the student stated above for the assignment
of the Programming course. I declare that this solution is my own work. I have not
copied or used third party solutions. I have not passed my solution to my classmates,
neither made it public.
Students' regulation of Eötvös Loránd University (ELTE Regulations Vol. II. 74/C.§)
states that as long as a student presents another student's work - or at least the
significant part of it - as his/her own performance, it will count as a disciplinary
fault. The most serious consequence of a disciplinary fault can be dismissal of the
student from the University.*/
```

When submitting on TMS upload the program with the output enabled that only prints the result of the specific subtask e.g.: *if you are solving a) do not include the outputs for tasks b), c), d)*!

Pirates Of The Looted Seas

The Caribbean seas are bustling with pirates hunting for riches. The Royal Navy has started tracking their activities, monitoring the treasures they've looted. Each record includes the value of the loot, the region where it was found, and the pirate's name. Some pirates have been active across multiple regions.

Input

The first line of input contains N (1 ≤ N ≤ 50), the number of recorded lootings, and T (1 ≤ T ≤ 100), a threshold for tasks (b) and (d). Each of the next N lines contains an integer L (1 ≤ L ≤ 100), representing the loot's value, (in thousands of doubloons), R a string representing the region name (e.g., Tortuga, PortRoyal, etc.), and P as the pirate's name.

Write a program that solves the following tasks:

Tasks:

a) Print the name of the pirate with the lowest value of single loot.
b) Count the number of records which loot value is greater than or equal to the given threshold. Output this count as a single number. If no pirate meets this criterion, output 0.
c) For each unique region, calculate the total loot value and print it along with the region's name. Maintain the order in which the regions first appear.
d) Find the number of the first record for Jack if his summation of loot's value is greater than the given threshold. Then print Jack's name and the total value of loot in a new line.

Output

Each task's output should be separated by new lines. Do *NOT* print the text in red; the red color is used here solely to illustrate and distinguish each subtask's output for clarity.

Example

Input

```
6 30
20 Tortuga Barbossa
40 Tortuga Jack
50 Havana Jack
25 Tortuga Will
30 Havana Barbossa
35 Tortuga Jack
```

Output

```
Task a):
Barbossa
Task b):
4
Task c):
Tortuga 120
Havana 80
Task d):
2
Jack 125
```